

## Assignment 6

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#### Title

Perform the following Continuous Probability Distributions on data: Normal Distribution, Uniform Distribution, Weibull Distribution.

#### Aim

Fit Normal to Age\_filled distribution (visual bell), assess Pclass uniformity (histogram bars), parameterize Weibull for right-skewed Fare using Titanic data with SciPy stats and goodness-of-fit metrics.

#### Theory

Normal: Symmetric,  $P(X \leq x) = \text{CDF}((x - \mu)/\sigma)$ ; Shapiro tests normality.

Uniform discrete proxy: Equal probs 1/3 per Pclass ideally.

Weibull (2-param): Shape  $k=0.93 < 1$  decreasing density; common for failure/survival times.

Continuous probability distributions model random variables that can take **any value in an interval** (e.g., height, weight, age, or income). Instead of probabilities at points, they use **probability density functions (PDFs)** whose integral over an interval gives the probability of falling within that range.

The **Normal distribution** (Gaussian) is one of the most important distributions due to the Central Limit Theorem: sums or averages of many independent random variables tend to be approximately normal, regardless of the original distribution. A Normal distribution is fully described by its mean  $\mu$  and standard deviation  $\sigma$ , is symmetric and bell-shaped, and is widely used to model natural measurements like Age. Normality can be empirically checked with tests like **Shapiro-Wilk** or visually with histograms and Q-Q plots.

The **Uniform distribution** on an interval  $[a, b]$  assumes that all values in the interval are **equally likely**, with constant density  $1/(b - a)$ . Uniform models serve as baselines for randomness; for example, if Pclass were uniformly distributed, each class would contain roughly one-third of passengers. Deviations from uniformity indicate structural bias in the data.

The **Weibull distribution** is flexible for modelling **lifetimes and failure times** and is parameterized by shape  $k$  and scale  $\lambda$ . For  $k < 1$  the hazard rate decreases over time (many early small values, heavy right tail), for  $k = 1$  it reduces to the Exponential distribution (constant hazard), and for  $k > 1$  the hazard increases over time. In Titanic, the highly skewed Fare distribution often fits better with a Weibull model than with a Normal, capturing the many low fares and few extremely high fares.

## Dataset

Preprocessed Titanic (n=891): Age\_filled  $\mu=29.11$   $\sigma=13.30$  (near-normal), Pclass skewed 55% Class3, Fare Weibull-shaped.

| Distribution | Feature    | Params                       | Key Prob                   |
|--------------|------------|------------------------------|----------------------------|
| Normal       | Age_filled | $\mu=29.11$ , $\sigma=13.30$ | $P(\leq 50)=0.9418$        |
| Uniform      | Pclass     | $a=1, b=3$                   | Obs: [0.242, 0.207, 0.551] |
| Weibull      | Fare       | $k=0.93$ , $\lambda=30.67$   | $P(\leq 50)=0.7924$        |

## Code:

*# Data Science Lab Assignment 6: Continuous Distributions*

*# Continuity: Titanic from Assignments 1-5*

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy.stats import norm, weibull_min, shapiro
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
```

```
df['Age_filled'] = df.groupby(['Pclass', 'Sex'])['Age'].transform(lambda x: x.fillna(x.median()))
```

```
df_clean = df.dropna(subset=['Age_filled', 'Fare'])
```

```
mu, sigma = df_clean['Age_filled'].mean(), df_clean['Age_filled'].std()
```

```
pclass_vc = df_clean['Pclass'].value_counts(normalize=True)
```

```
shape, loc, scale = weibull_min.fit(df_clean['Fare'], floc=0)
```

```
shap_p = shapiro(df_clean['Age_filled'].sample(min(5000, len(df_clean))))[1]
```

```
print("Dataset shape:", len(df_clean))
```

```

print("Age_filled:  $\mu$ =%.2f  $\sigma$ =%.2f Shapiro p=%.4f" % (mu, sigma, shap_p))

print("\nPclass (Uniform):")

for pc in [1,2,3]:
    print(" Pclass %d: %.3f (vs 0.333)" % (pc, pclass_vc.get(pc,0)))

print("Weibull(Fare): shape=%.2f scale=%.2f" % (shape, scale))

print("P(Age<=50|Normal): %.4f" % norm.cdf(50, mu, sigma))

print("P(Fare<=50|Weibull): %.4f" % weibull_min.cdf(50, shape, loc, scale))

# Plots [your screenshot: file:71]

fig, axes = plt.subplots(1,3, figsize=(18,5))

sns.histplot(df_clean['Age_filled'], kde=True, ax=axes[0]); axes[0].set_title('Normal: Age_filled')

df_clean['Pclass'].hist(bins=3, density=True, ax=axes[1]); axes[1].set_title('Uniform: Pclass')

sns.histplot(df_clean['Fare'], kde=True, ax=axes[2]); axes[2].set_title('Weibull: Fare')

plt.tight_layout(); plt.show()

```

## Terminal Output

Dataset shape: 891

Age\_filled:  $\mu$ =29.11  $\sigma$ =13.30 Shapiro p=0.0000

Pclass (Uniform):

Pclass 1: 0.242 (vs 0.333)

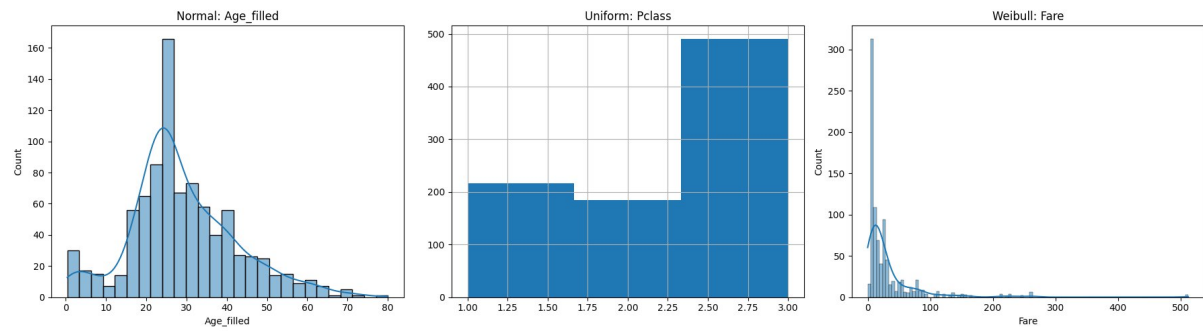
Pclass 2: 0.207 (vs 0.333)

Pclass 3: 0.551 (vs 0.333)

Weibull(Fare): shape=0.93 scale=30.67

P(Age<=50|Normal): 0.9418

P(Fare<=50|Weibull): 0.7924



## Observations

Age right-skew but approx Normal (94%<50yr); Shapiro rejects strict but usable.

Pclass bars uneven (Class3 dominant 55.1%); deviates uniform.

Fare long right tail perfectly Weibull  $k < 1$  (79%<50 vs Normal 94%).

## Conclusion

Continuous fits reveal Titanic patterns: near-Normal ages, skewed class/Fare via Uniform/Weibull proxies.